

Early menopause and risk of osteoporosis, fracture and mortality: a 34-year prospective observational study in 390 women

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Accepted 26 February 2012. Published Online 25 April 2012.

Objective A prospective evaluation of the long-term effects of early menopause on mortality, risk of fragility fracture and osteoporosis.

Design Prospective population-based observational study.

Setting Malmö, Sweden.

Population A total of 390 white north European women aged 48 years at the start of the study.

Methods At baseline, bone mineral density (BMD) was measured by single-photon absorptiometry (SPA) in the distal forearm and menopausal status was noted. Menopause was determined according to the World Health Organization criterion of a minimum of 12 months of continuous amenorrhoea. Women were divided into early menopause (occurring before age 47 years) and late menopause (occurring at age 47 years or later). At age 77, forearm BMD was re-measured by SPA and proximal femur and lumbar spine BMD were measured by dual-energy X-ray

absorptiometry (DXA). The prevalence of osteoporosis was determined using the DXA data. Mortality rate and the incidence of fractures were registered up until age 82. Data are presented as means with 95% confidence intervals (95% CI).

Main outcome measures Incidence of fragility fractures, mortality, prevalence of osteoporosis at age 77.

Results Women with early menopause had a risk ratio of 1.83 (95% CI 1.22–2.74) for osteoporosis at age 77, a risk ratio of 1.68 (95% CI 1.05–2.57) for fragility fracture and a mortality risk of 1.59 (95% CI 1.04–2.36).

Conclusions Menopause before age 47 is associated with increased mortality risk and increased risk of sustaining fragility fractures and of osteoporosis at age 77.

Keywords Bone mass, fractures, menopause, mortality, osteoporosis.

Please cite this paper as: Svejme O, Ahlborg H, Nilsson J, Karlsson M. Early menopause and risk of osteoporosis, fracture and mortality: a 34-year prospective observational study in 390 women. BJOG 2012;119:810–816.

Introduction

Published literature suggests that an early menopause can predict osteoporosis and its clinical manifestation, fragility fractures.^{1–6} However, opinions differ, because some long-term studies have concluded that age at menopause does not influence bone mineral density (BMD) after the age of 70 years.^{7–9} Furthermore, most studies on the subject have been cross-sectional rather than population-based cohorts and have used retrospective definitions of menopause age obtained through interviews or questionnaires—a study design subject to recall bias. In addition, most studies have evaluated outcome in the first decade following the menopause,¹ whereas the risk of fracture only starts to increase exponentially several decades after the menopause,¹⁰ and

the mean age for a female hip fracture in Sweden is currently 82 years.¹⁰

An ideal study to estimate the long-term impact of menopause on the prevalence of osteoporosis and fracture risk should follow a homogeneous population of women from the perimenopausal period through several decades of life until the participants reach the age when osteoporosis and fragility fractures become a common problem.

The Malmö Perimenopausal Study is one such population-based prospective observational study where women have been followed from age 48 years onwards. The 16-year follow-up data have been presented,⁸ and the follow up in the current report is extended to a total of 29 years for the BMD measurements and 34 years for the fracture and mortality evaluations.

In this study we hypothesise that an early menopause is a risk factor for osteoporosis, mortality and fragility fractures in a long-term prospective study.

Methods

In 1977, 390 women aged 48 years were recruited to this prospective observational study, originally with the aim of evaluating changes in BMD and endocrine parameters in the perimenopausal and postmenopausal periods.^{11,12} All white north European female residents of the city of Malmö, Sweden who were born during the latter half of 1929 were selected from the city population records.¹¹ At age 77 years, all eligible participants were re-invited for BMD measurements. At this point, 298 women were still alive and 92 had died. There was no information available on cause of death. One hundred of the 298 women still alive had relocated or declined further participation because of disease or for personal reasons, leaving 198 women to attend the follow-up measurement 29 years after baseline. Lifestyle parameters, medical conditions, medications used and gynaecological history were noted at baseline and at age 77. At baseline, the women were asked specifically whether they were still menstruating or not. We then used the WHO definition that requires 12 months of continuous amenorrhoea to define menopause.¹³ Based on this information we were able to divide the women into two categories; an early menopause group consisting of women in whom the menopause occurred before age 47 and a late menopause group including women in whom the menopause occurred at age 47 or later. The mean age at menopause in the early menopause group was 42 years, compared with the average age at menopause of 51 years in the western world.^{14–18}

Bone mineral density was measured by single-photon absorptiometry (SPA), which gauges BMD at one mainly trabecular, and one dominantly cortical site of the forearm, as described by Naucler et al.¹⁹ The SPA measurements were performed both at age 48 and at age 77. The prospective evaluation of bone mass therefore spanned 29 years. The same apparatus and the same anatomical site were used and all data from both measurements were analysed by the same technician. Measurements of a standardised phantom every second week all through the study period did not reveal any long-term drift of the densitometer. At age 77, BMD (g/cm^2) was also measured by dual energy X-ray absorptiometry (DXA; GE Lunar Corp., Madison, WI, USA; Lunar Prodigy®) in the hip and lumbar spine. The precision of the BMD measurements ranged from 0.5 to 3%, depending on application.²⁰ Osteoporosis was defined according to the WHO definition, i.e. -2.5 standard deviation (SD) from the reference value in young adults.²¹ In absolute numbers, these cutoff values were at

$0.706 \text{ g}/\text{cm}^2$ for hip BMD and $0.907 \text{ g}/\text{cm}^2$ for lumbar spine BMD. Height and weight were measured by standard equipment at both 48 and 77 years.

We identified fractures that occurred in these 390 women from repeated searches of hospital registrations and digitised databases, from age 48 years until death, relocation or until the endpoint date 30 September 2011 (age 82 years). The fracture registration period therefore spanned 34 years, and as the incident fractures were objectively registered through their hospital attendances, all women contributed data, even those who declined follow-up BMD measurements. Fragility fractures were defined as low-energy fractures (a fall on the same level) of the wrist, proximal humerus, spine, hip, pelvis, tibia condyle and ankle. Nonclinical asymptomatic vertebral fractures and nonosteoporotic high-energy fractures were not included in our calculations. This fracture ascertainment method has been thoroughly evaluated and used in numerous epidemiological fracture studies.^{22–24} Everyone in the Malmö region attends the same trauma unit because there is only one emergency department in the city, and all radiographic examinations were routinely examined by two radiologists and then registered and indexed. Radiographs and reports have been kept on file for each person since the beginning of the twentieth century. Additionally, residents of Malmö who sustain fractures in other geographical regions are referred to the Orthopaedic Department of Skåne University Hospital in Malmö for follow up, at which the fracture is classified in the records as low or high energy. Fewer than 3% of all fracture patients in the city visit a private physician. Most of these injuries would have been minor fractures not necessitating treatment, such as non-dislocated digit fractures.^{22–24} Classified fractures were verified by the radiologists' original reports. Nine participants who had relocated to other regions in Sweden were telephoned and asked whether they had sustained any fracture. If fractures were reported, they were verified through case reports acquired from the respective hospital. In 11 women, fracture records could not be obtained because these women had either died after having relocated or could not be located. In these cases, we used the relocation date as the endpoint data in the risk calculations. Mortality data during the 34-year follow-up period was provided from the national population registers. A flow-chart following the 390 women until their last BMD measurement and the end of fracture and mortality registration is shown in Figure 1.

Approval was obtained from the Ethics Committee of Lund University and the study was conducted in accordance with the norms of the Helsinki Declaration of 2001. Written informed consent for collection of data was obtained from each individual. The technical equipment was validated by the Swedish Radiation Protection Inspectorate and by the hospital's own radiation protection

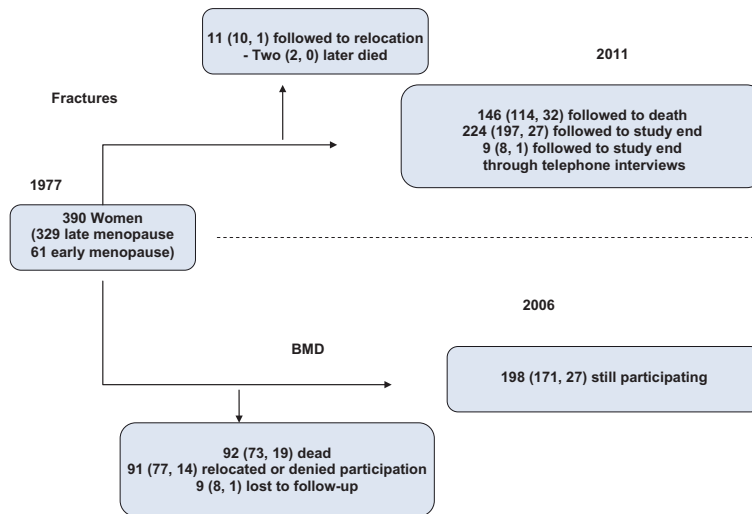


Figure 1. Flow-chart of the participants, 390 at study start, until the last BMD measurement in 2006 and the fracture registration until 30 September 2011. At the last BMD measurement in 2006, 198 of the 298 women still alive (66%) attended. Fracture registration from study start until 30 September 2011 was complete in 379 women (97%), in 370 cases through the hospital archives and the regional databases and in nine women through telephone interviews. Within brackets are numbers for the late menopause group and the early menopause group, respectively.

committee. The Swedish Data Inspection Board approved both the data collection and the database. Statistical processing was carried out using STATISTICA software version 7.1 (StatSoft, Milton Keynes, UK). Data are shown as means with 95% confidence intervals (95% CI). Group comparisons between women with early and late menopause were performed using chi-square tests, log-rank tests, risk ratio and rate ratio calculations and the Student's *t* test between means. Fracture incidence and mortality rate in the two cohorts following baseline were calculated taking person-years into account and are presented with Kaplan–Meier survival curves.

Results

At age 48, no group differences were seen with regards to age at menarche or anthropometrics, whereas distal forearm BMD was 0.43 SD (95% CI 0.14–0.72) lower in the early menopause group (Table 1).

At age 77, 56% (15/27) of women with early menopause had osteoporosis, in comparison with 30% (52/171) of women with late menopause ($P = 0.01$), resulting in a risk ratio of 1.83 (95% 1.22–2.74) (Table 2).

During the 34-year fracture follow-up period, 33% (128/390) of the women had sustained at least one fragility fracture (Table 2). In the early menopause group, the fracture incidence per 10³ person-years was 19.45 compared with 11.60 in the late menopause group, giving a risk ratio of 1.68 (95% CI 1.05–2.57) for sustaining a fragility fracture (Table 2, Figure 2).

Mortality rate was 52.4% (32/61) in the early menopause group and 35.2% (116/329) in the late menopause group,

giving a relative mortality risk of 1.59 (95% 1.04–2.36) (Table 2, Figure 3).

Discussion

The results of this population-based prospective observational study demonstrate that an early menopause is a significant risk factor for osteoporosis, fragility fracture and mortality. There is an abundance of surveys associating early menopause with osteoporosis and risk of fracture. However, the majority of those studies have been cross-sectional and ascertained menopausal age retrospectively.^{1,4–6,25–27} To our knowledge, this is the first report with a prospective study design and a follow-up period of more than three decades. In addition, we not only included osteoporosis as the surrogate endpoint marker, but also ascertained the clinically relevant endpoints; fracture and mortality.

Previous reports have generally been robust in establishing the association of early menopause with the presence of low BMD and fragility fractures during the first postmenopausal decade, whereas this relationship seems to disappear with increasing age.¹ Our study was designed to evaluate the influence of age at menopause on the risk of osteoporosis as well as fracture and mortality.^{1,4,5,26,27} Our results corroborate most of the previously published data. In one review in 2007 that included predominantly cross-sectional studies and prospective short-term studies, Gallagher¹ concluded that most reports support the association of early menopause with the risk of fracture and future osteoporosis. Gardsell et al.² followed 733 women for 11 years and were able to establish a relationship between early menopause and increased fracture incidence but only up to age

Table 1. Characteristics of the 390 women of the original cohort when evaluated at study start at age 48 years

Variable	Early menopause (n = 61)	Late menopause (n = 329)	P value*
Age (years)	48.3 (48.3–48.3)	48.3 (48.3–48.3)	–
Menarche (years)	13.9 (13.4–14.3)	14.0 (13.9–14.2)	0.45
Height (cm)	163.8 (162.3–165.3)	164.1 (163.5–164.6)	0.78
Weight (kg)	63.2 (60.8–65.8)	63.5 (62.4–64.6)	0.88
Forearm BMD (g/cm²)	0.52 (0.50–0.54)	0.55 (0.54–0.55)	0.002
Menopausal age (years)	42.1 (40.8–43.4)	N/A	
History of breastfeeding			
Yes	42 (69%)	260 (79%)	0.05
No	18 (30%)	60 (18%)	
Missing data	1 (1%)	9 (3%)	
Children			
0	15 (25%)	37 (11%)	0.31
1–3	40 (65%)	271 (82%)	
>3	6 (10%)	20 (6%)	
Missing data	0	1	
Current physical activity			
High	12 (20%)	96 (29%)	0.13
Low	49 (80%)	233 (71%)	
Current smoking			
Yes	33 (54%)	153 (46%)	0.14
No	22 (36%)	158 (48%)	
Missing data	6 (10%)	18 (6%)	
History of oral contraceptives			
Yes	10 (16%)	86 (26%)	0.10
No	51 (84%)	243 (74%)	
Current calcium intake			
<400 mg/day	14 (23%)	56 (17%)	0.27
≥400 mg/day	47 (77%)	273 (83%)	

Data presented as mean with 95% CI or as numbers with proportion (%).

BMD was measured by SPA at the distal radius. Chi-square tests and Student's *t* test were used for *P* value calculation.

**P* value for early menopause versus late menopause.

70 years. Johansson and Mellstrom,²⁵ in a cross-sectional retrospective study of 7549 women from six different birth cohorts, reported that fracture risk in each cohort was obviously associated with menopause occurring before the age of 49. However, in contrast to the present study, both menopause age and fracture incidence were estimated retrospectively through questionnaires and high-energy-related fractures were also included in the calculations. In a study of 555 Californian women aged 60–89 years and with 23–34 years of postmenopausal period, Kritz-Silverstein and Barrett-Connor⁶ found lower BMD in women with menopause before age 48. However, no fracture data were included in this report. Furthermore, few have evaluated whether an early menopause influences the level of BMD or fracture incidence decades after the menopause, and the sparse published data suggest a fading impact of early menopause with increasing age.¹ This may be caused by the fact that the rapid estrogen-associated bone loss in the first postmenopausal decade is replaced by a slower age-related loss in BMD and that an increasing number of other risk

factors for low BMD and fracture appear in older women, obscuring the effect of early menopause.^{2,7–9}

The conflicting results in the literature may be explained by the use of different cutoff ages for early menopause. With a lowering of the threshold when defining early menopause, it seems as if the association between age at menopause and risk of osteoporosis and fragility fracture is strengthened.^{4–6,26,27} A cross-sectional study including 1050 Argentinean women aged 50–88 years suggested that women who reached the menopause before age 45 had lower BMD than women with later menopause, and half of the 49 women who had sustained a hip fracture were in the early menopause group.²⁷ A French cross-sectional study of 1667 women reported that women in whom the menopause had occurred before age 40, two decades later had lower BMD than women with higher menopause age.⁵ Van der Voort et al.,⁴ in one cross-sectional evaluation involving 4725 women, concluded that a menopause before the age of 45 was associated with a higher fracture risk above age 70 but lower BMD only up to the age of 65.

Table 2. Characteristics of the 198 women who were still participating in the measurements at study end

Variable	Early menopause group	Late menopause group	P value*
Age (years)	76.8 (76.6–77.1)	76.5 (76.4–76.6)	–
Age at menopause (years)	42.1 (40.8–43.4)	51.0 (50.7–51.4)	<0.001
Height (cm)	160.3 (157.9–162.6)	160.7 (159.9–161.7)	0.66
Weight (kg)	67.5 (62.9–72.1)	67.3 (65.5–69.2)	0.96
BMD forearm (g/cm²)	0.36 (0.32–0.40)	0.37 (0.36–0.38)	0.58
BMD hip (g/cm²)	0.79 (0.74–0.84)	0.83 (0.80–0.85)	0.23
BMD lumbar (g/cm²)	0.98 (0.90–1.07)	1.04 (1.01–1.07)	0.19
History of hormone replacement therapy			
Yes	6 (22%)	16 (10%)	0.05
No	21 (78%)	155 (90%)	
History of bisphosphonates			
Yes	2 (7%)	20 (12%)	0.51
No	25 (93%)	151 (88%)	
History of oral corticosteroids			
Yes	1 (4%)	10 (6%)	0.65
No	26 (96%)	161 (94%)	
Deceased			
Yes	32 (52%)	116 (35%)	0.01
No	29 (48%)	213 (65%)	
Person-years	1719	10 007	
Osteoporosis			
Yes	15 (56%)	52 (30%)	0.01
No	12 (44%)	119 (70%)	
Fragility fractures			
Yes	27 (44%)	101 (31%)	0.04
No	34 (56%)	228 (69%)	
Person-years	1389	8719	
Distal radius fractures			
Yes	11 (18%)	53 (16%)	0.71
No	50 (82%)	276 (84%)	
Person-years	1583	9074	

Data are presented as mean with 95% CI or as numbers with proportion (%). BMD was measured by SPA at the distal radius ($n = 176$) and by DXA in total hip ($n = 194$) and lumbar spine ($n = 197$).

Osteoporosis was classified according to the WHO classification¹⁴ as a BMD T -score below 2.5 SD measured by DXA.

Fracture data are gathered from the 379 women available for fracture follow up.

Chi-square tests and Student's t test were used for P value calculation.

* P value for early menopause versus late menopause.

Finally, the Rotterdam study²⁶ inferred a relative risk of incident vertebral fractures of 2.7 in women with menopause before age 46 but a relative risk of only 1.3 in women with menopause between age 46 and 50, in comparison with women who had menopause after age 50. In summary, these reports indicate a dose–response relationship between age at menopause and fracture risk, i.e. the earlier the onset, the more sustained the impact.

There are also studies that oppose an association between early menopause and osteoporosis and increased fracture risk. In a cross-sectional report by Gerdhem and Obrant,⁷ 1044 75-year-old women were asked about their age at menopause, which did not correlate with BMD at age 75. The same conclusion was drawn by Francucci

et al.,⁹ who studied 782 women and reported that menopause before age 44 was associated with lower BMD up to age 55 but not beyond, indicating a fading effect of age at menopause on BMD with advancing age. In contrast to these cross-sectional evaluations, our prospective data illustrate that the effect of early menopause results in a significantly higher prevalence of osteoporosis at age 77 in women with early menopause as well as a higher incidence of fractures. The discrepancy in study conclusions could be the result of different study designs, the different strategies of estimating age at menopause and the different definitions of early and late menopause.

The reason for the higher fracture risk among women with early menopause can only be speculated upon. One of

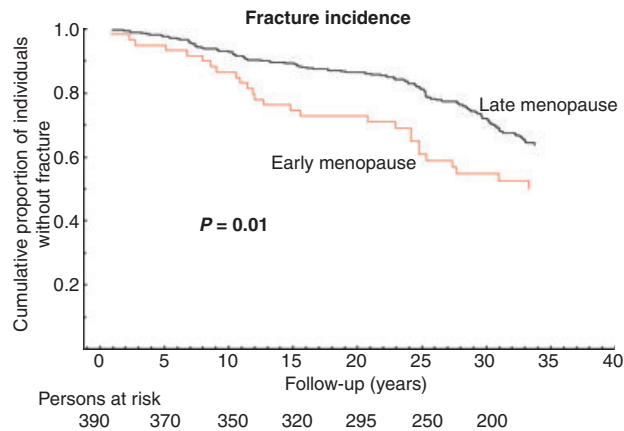


Figure 2. Kaplan–Meier curve presenting fracture incidence in the two groups during the follow-up period. Remaining individuals at risk are provided along the time axis. *P* value is calculated through log-rank test. Rate ratio for fragility fracture incidence taking person years into account was 1.68 (95% CI 1.05–2.67).

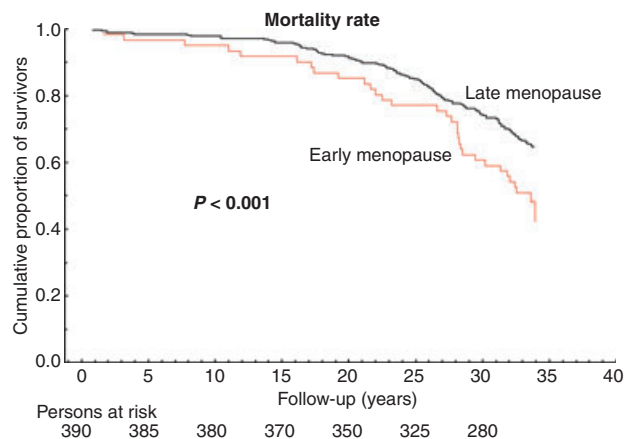


Figure 3. Kaplan–Meier curve presenting mortality rate in the two groups during the follow-up period. Remaining individuals at risk are provided along the time axis. *P* value is calculated through log-rank test. Relative mortality risk taking person-years into account was 1.59 (95% CI 1.04–2.36).

the most recognised risk factors for fractures is low BMD. It is well established in prospective epidemiological studies that a decrease of 1.0 SD increases fracture risk by 50%.²⁸ The higher fracture risk in women with early menopause in the current study is probably to some extent mediated by a lower BMD, as the early menopause cohort had a significantly higher risk of osteoporosis at age 77 and already, at age 48 years, had on average a 0.4 SD lower BMD than those with late menopause. However, the lower BMD level does not seem to be entirely able to explain the increased fracture risk. We must therefore speculate as to whether factors beyond bone mass, such as inferior muscle strength or inferior neuromuscular function, ought to be found in women with early menopause.

Any possible relationship between the high mortality risk in women with early menopause can likewise only be a matter of speculation. It has been suggested that the higher mortality rate in women with an early menopause is mediated through higher co-morbidity²⁹ and effects dependent on the hormonal transition associated with menopause.²⁹ A higher fracture incidence in women with an early menopause could be associated with increased fracture-related mortality,^{30–32} but the differences in mortality risk could also be related to group differences in general diseases, medication, nutritional intake, smoking and alcohol habits, level of physical activity and other lifestyle factors, all factors that have been reported to be associated with mortality risk.

The strengths of this study include the population-based study design, the homogeneity with regard to age and ethnicity, the 97% participation rate in the fracture and mortality evaluation and the—to our knowledge—unprecedented study length. Furthermore, prospective fracture registration through a well-validated method that only includes objectively verified fractures is likely to increase the reliability of our findings. The definition of the menopause using the World Health Organization (WHO) classification¹³ and determination of the age at menopause being established at the start of the study, instead of using retrospectively estimated age at the menopause as in most previous studies, must also be regarded as a study strength. Study limitations include the sample size and the number of drop-outs in the bone mass evaluations. As the participation rate in the bone mass evaluation was only 51% (198/390) these data must be regarded as less reliable than the fracture and mortality data. Nevertheless, a participation rate in the BMD measurements of 66% among the women still alive is creditable three decades after the start of the study. It would also have been of interest to evaluate an even lower cutoff value of age at menopause in our study. However, this would have to be based on retrospective estimations of menopause age, with the risk of recall bias. Instead we chose to use only data provided at the actual date of the evaluation and then applied the generally accepted WHO classification when defining age at menopause, after which the women were divided into groups of early and late menopause. It would also have been advantageous to have included preplanned spine radiographs in the evaluation, as many women with osteoporotic spinal fractures never seek medical advice at the time of the fracture.

Considering the restrictions discussed above, we can conclude that a menopause before age 47 is associated with an increased risk of mortality, fragility fractures and osteoporosis at age 77.

Disclosure of interests

None.

Contribution to authorship

OS and MKK were the main authors and were responsible for study design and statistical calculations; MKK also collected the data. HJA was responsible for study design and data collection and co-authored the paper. JÅN performed the statistical calculations.

Details of ethics approval

At the start of the study in 1977, no permission from the institutional review board and no consent form were required; the women were asked to provide oral informed consent. However, later in the course of the study, permission was granted by the ethics committee of the University of Lund, the parent organisation of Skåne University Hospital. Approval date was 25 January 2000 and reference number is LU 625-99.

Funding

None.

Acknowledgements

None. ■

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